

FULL DETAILS (Read-only) -> [Click Here to Create PDF for Current Dataset of Trial](#)

CTRI Number	CTRI/2022/02/040169 [Registered on: 10/02/2022] Trial Registered Prospectively															
Last Modified On:	27/02/2023															
Post Graduate Thesis	Yes															
Type of Trial	Observational															
Type of Study	Cross Sectional Study															
Study Design	Other															
Public Title of Study	Expression of HMGB1 and RAGE In patients with gum related diseases with and without type 2 diabetes mellitus															
Scientific Title of Study	EXPRESSION OF HIGH MOBILITY GROUP BOX 1 (HMGB1) AND RECEPTOR FOR ADVANCED GLYCATION END PRODUCTS (RAGE) IN GINGIVAL TISSUES OF PERIODONTITIS PATIENTS WITH AND WITHOUT TYPE 2 DIABETES MELLITUS															
Trial Acronym																
Secondary IDs if Any	<table border="1"> <thead> <tr> <th>Secondary ID</th><th>Identifier</th></tr> </thead> <tbody> <tr> <td>NIL</td><td>NIL</td></tr> </tbody> </table>	Secondary ID	Identifier	NIL	NIL											
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Details of Principal Investigator or overall Trial Coordinator (multi-center study) Modification(s)	<table border="1"> <tr> <td>Name</td><td>R Ramya</td></tr> <tr> <td>Designation</td><td>POST GRADUATE</td></tr> <tr> <td>Affiliation</td><td>SRM DENTAL COLLOGE</td></tr> <tr> <td>Address</td><td>Bharati salai,ramapuram Coast guard sailors married accommodations, GM Pettai road , Royapuram, chennai-600013 Chennai TAMIL NADU 600089 India</td></tr> <tr> <td>Phone</td><td>08903929674</td></tr> <tr> <td>Fax</td><td></td></tr> <tr> <td>Email</td><td>ramya7raj@gmail.com</td></tr> </table>		Name	R Ramya	Designation	POST GRADUATE	Affiliation	SRM DENTAL COLLOGE	Address	Bharati salai,ramapuram Coast guard sailors married accommodations, GM Pettai road , Royapuram, chennai-600013 Chennai TAMIL NADU 600089 India	Phone	08903929674	Fax		Email	ramya7raj@gmail.com
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Details of Contact Person Scientific Query	<table border="1"> <tr> <td>Name</td><td>Dr Devapriya AM</td></tr> <tr> <td>Designation</td><td>Professor</td></tr> <tr> <td>Affiliation</td><td>SRM DENTAL COLLOGE</td></tr> <tr> <td>Address</td><td>ramapuram Bharati salai,ramapuram Chennai TAMIL NADU 600089 India</td></tr> <tr> <td>Phone</td><td>09840197121</td></tr> <tr> <td>Fax</td><td></td></tr> <tr> <td>Email</td><td>devapriyams@gmail.com</td></tr> </table>		Name	Dr Devapriya AM	Designation	Professor	Affiliation	SRM DENTAL COLLOGE	Address	ramapuram Bharati salai,ramapuram Chennai TAMIL NADU 600089 India	Phone	09840197121	Fax		Email	devapriyams@gmail.com
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	Email		devapriyamds@gmail.com	
Source of Monetary or Material Support		R.Ramya, Self funded study		
Primary Sponsor	Name	R Ramya		
	Address	C-2/4,coast guard quarters ,GM Pettai road, Royapuram , Chennai-600013		
	Type of Sponsor	Other [Self Funded]		
Details of Secondary Sponsor	Name		Address	
	NIL		NIL	
Countries of Recruitment		India		
Sites of Study Modification(s)	No of Sites = 1			
	Name of Principal Investigator	Name of Site	Site Address	Phone/Fax/Email
	R RAMYA	SRM dental College	Department of Periodontics, Bharati salai, Ramapuram Chennai TAMIL NADU	08903929674 ramya7raj@gmail.com
Details of Ethics Committee Modification(s)	No of Ethics Committees= 1			
	Name of Committee			Approval Status
INSTITUTIONAL REVIEW BOARD,SRM DENTAL COLLEGE			Approved	
Regulatory Clearance Status from DCGI	Status			
	Not Applicable			
Health Condition / Problems Studied	Health Type	Condition		
	Patients	(1) ICD-10 Condition: E119 Type 2 diabetes mellitus without complications,		
Intervention / Comparator Agent	Type		Name	Details

	Age From	18.00 Year(s)	
	Age To	75.00 Year(s)	
	Gender	Both	
	Inclusion Criteria	Details INCLUSION CRITERIA FOR SYSTEMICALLY AND PERIODONTALLY HEALTHY GINGIVAL TISSUE SAMPLING - â€¢Age group years â‰¥ 18years - â€¢Good systemic health - â€¢Presence of at least 20 natural teeth excluding 3rd molars - â€¢< 10% bleeding sites - â€¢Probing depth â‰¥ 3mm - â€¢No clinical attachment loss INCLUSION CRITERIA FOR PERIODONTALLY DISEASED TISSUE SAMPLING - â€¢Patients with age group â‰¥ 18years - â€¢Patients with stage III/IV periodontitis (according to 2017 EPF/AAP classification) - â€¢FMPS â‰¥ 20% - â€¢FMBPS â‰¥ 20% - â€¢Periodontal probing depth â‰¥ 6mm - â€¢Clinical attachment loss â‰¥ 5mm - â€¢Radiographic evidence of bone loss for stage III/IV bone loss extending to middle or apical third of root. INCLUSION CRITERIA FOR PERIODONTALLY DISEASED WITH TYPE 2 DIABETES MELLITUS TISSUE SAMPLING - â€¢Patients with age group â‰¥ 18years - â€¢Patients with stage III/IV periodontitis grade C (according to 2017 EPF/AAP classification) - â€¢FMPS â‰¥ 20% - â€¢FMBPS â‰¥ 20% - â€¢Periodontal probing depth â‰¥ 6mm - â€¢Clinical attachment loss â‰¥ 5mm - â€¢Radiographic evidence of bone loss for stage III/IV bone loss extending to middle or apical third of root. - â€¢HbA1c â‰¥ 7% diagnosed and under medication for the past more than 1 year	
	ExclusionCriteria	Details EXCLUSION CRITERIA - â€¢Patients with any other systemic diseases (other than T2DM)/ auto immune diseases - â€¢Immunosuppressants, antibiotics, anti-inflammatory drugs usage in the last six months - â€¢Pregnant or lactating mothers. - â€¢Smokers (current and past) - â€¢Previous history of periodontal therapy	
	Method of Generating Random Sequence	Not Applicable	
	Method of Concealment	Not Applicable	
	Blinding/ Masking	Not Applicable	
	Primary Outcome	Outcome If an increased or over-expression of both HMGB1 and RAGE can be demonstrated as hypothesized, it will possibly improve our understanding on the role of HMGB1-RAGE axis in the pathogenesis of diabetes associated periodontal disease	TimePoints Baseline

Secondary Outcome	<table> <tr> <th>Outcome</th><th>TimePoints</th></tr> <tr> <td> HMGB1 can possibly be used as a diagnostic biomarker Identifying the role of HMGB1 and RAGE as key mediators of inflammation can enable novel therapeutic drug based interventions based on blocking the HMGB1-RAGE axis that could possibly mitigate the inflammation mediated tissue destruction in diabetes associated periodontitis . </td><td>14-18 months</td></tr> </table>	Outcome	TimePoints	HMGB1 can possibly be used as a diagnostic biomarker Identifying the role of HMGB1 and RAGE as key mediators of inflammation can enable novel therapeutic drug based interventions based on blocking the HMGB1-RAGE axis that could possibly mitigate the inflammation mediated tissue destruction in diabetes associated periodontitis .	14-18 months
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Target Sample Size	Total Sample Size="33" Sample Size from India="33" Final Enrollment numbers achieved (Total)= "33" Final Enrollment numbers achieved (India)="33"				
Phase of Trial	N/A				
Date of First Enrollment (India)	11/02/2022				
Date of Study Completion (India)	11/05/2022				
Date of First Enrollment (Global)	Date Missing				
Date of Study Completion (Global)	Date Missing				
Estimated Duration of Trial	Years="1" Months="4" Days="10"				
Recruitment Status of Trial (Global) Modification(s)	Not Applicable				
Recruitment Status of Trial (India)	Completed				
Publication Details Modification(s)	yet to be published				
Individual Participant Data (IPD) Sharing Statement	Will individual participant data (IPD) be shared publicly (including data dictionaries)? Response - NO				
Brief Summary	 ■HMGB1-RAGE binding results in acceleration and amplification of chronic inflammation via NF-kB.The binding promotes leukocyte and monocyte migration into the inflammatory milieu, endothelial dysfunction, promotes autophagy, Neutrophil NETosis , differentiation and maturation of dendritic				

cells, activates Th 17 cells, regulates B cell activation, activates inducible Nitric oxide synthase “sustaining and propagating the inflammatory injury

- ⌚ ■Till recently, RAGE to Advanced glycation end products (AGE) binding was principally implicated in the pathogenesis of periodontal destruction in DM. However, currently based on the emerging role of HMGB1-RAGE in the pathogenesis of chronic inflammatory diseases, it is imperative to investigate their expression in diabetes associated periodontal disease.
- ⌚ ■HMGB1 has not been evaluated in human gingival tissues from T2DM associated periodontal disease. Further the expression of RAGE and HMGB1 has not been evaluated collectively in T2DM associated periodontal disease from human gingival tissues.
- ⌚ ■Our study hypothesises that the levels of both HMGB1 and RAGE will be upregulated in T2DM associated periodontal disease when compared to periodontal healthy controls.

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